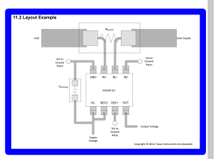
[5] INA241 - Phase Sensing

The INA240 device is a voltage-output, current-sense amplifier with enhanced PWM rejection that can sense drops across shunt resistors over a wide common-mode voltage range from -4 V to 80 V, independent of the supply voltage. The negative common-mode voltage allows the device to operate below ground, accommodating the flyback period of typical solenoid applications. Enhanced PWM rejection provides high levels of suppression for large common-mode transients (ΔV/Δt) in systems that use pulse width modulation (PWM) signals (such as motor drives and solenoid control systems). This feature allows for accurate current measurements without large transients and associated recovery ripple on the output voltage.
This device operates from a single 2.7-V to 5.5-V power supply, drawing a maximum of 2.4 mA of supply current. Four fixed gains are available: 20 V/V, 50 V/V, 100 V/V, and 200 V/V. The low offset of the zero-drift architecture enables current sensing with maximum drops across the shunt as low as 10-mV full-scale. All versions are specified over the extended operating temperature range (-40°C to +125°C), and are offered in an 8-pin TSSOP and 8-pin SOIC packages.

DESIGN NOTE:
Gain A1->A5 (INA241)
INA241A1 : 10V/V
INA241A2 : 20V/V (fitted)
INA241A3 : 50V/V
INA241A4 : 100V/V
INA241A5 : 200V/V

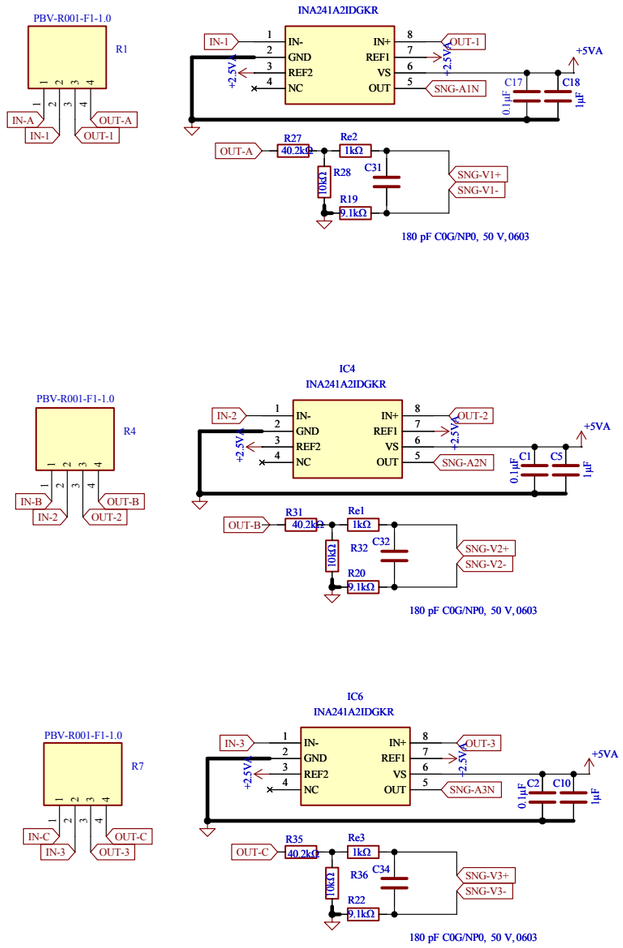
DESIGN NOTE:
Capacitor Découplage
0603
Ceramique X7R
>16V



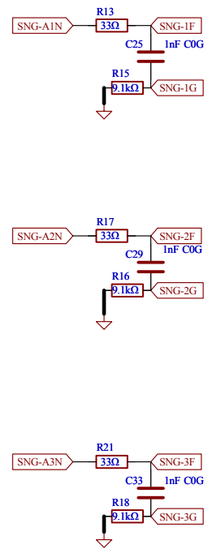
Description

Vout≈Vref+(Gain)×(I×Rshunt)

Shouse your INA240A...



INA241A + Shunt



FILTRE

Test Point

Comments: **References:** Flexible I/O worked examples; Flexible I/O source configuration	Company: OPEN THRUST LAB		Variant:	Git Hash: 7703dc5
	Board Name: PHASETRACKER		Project Name: OPEN_PHASE	
Sheet Title: INA241 - Phase Sensing	File Name: INA241 - Phase Sensing.kicad_sch	Designer: SIEBERT DIMITRY	Date: 2026-10-05	Revision: + (Unreleased)
Sheet Path: /Project Architecture/INA241_Phase_Sense/		Reviewer:	Size: A3	Sheet: 5 of 10